

US#5 Calendar Sync & US#6 Itinerary Map

Scope Note — Outlook Calendar

Outlook integration is not available in this sprint. Microsoft Azure App Registration requires a paid organizational account, which is not available for this project. All Outlook-related acceptance criteria (AC1, AC2, AC3, AC6, AC7, AC8) are deferred. The test cases below cover Google Calendar only.

Prerequisites

- App running at <http://localhost:3000>
 - A Google account added as a test user in the Google Cloud OAuth consent screen
 - GOOGLE_CALENDAR_CLIENT_ID, GOOGLE_CALENDAR_CLIENT_SECRET, TOKEN_ENCRYPTION_KEY, and NEXT_PUBLIC_GOOGLE_MAPS_API_KEY set in .env.local
 - Two test accounts: User A (group leader) and User B (group member)
 - One travel group with both users as members and at least two itinerary events, at least one with a location string (e.g. "Purdue University, West Lafayette, IN") and one without
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US#5 — Link External Calendar (Google)

TC-5-1: Link a Google Calendar account (AC1)

- Log in as User A
- Navigate to the group then click Calendar in the tab bar
- Click Connect Google Calendar
- Complete the Google OAuth consent screen (select account, grant permissions)
Expected: Redirected back to the app. The External Calendars panel shows the Google account as linked with the account email displayed. No error message is shown.

TC-5-2: Outlook Calendar — deferred (AC1 partial)

- Outlook is not configured. Clicking Connect Outlook Calendar (if shown) should either display a "not configured" message or redirect to a non-functional OAuth page. No further Outlook testing is performed this sprint.

TC-5-3: Select a calendar and enable sync (AC3, AC6)

- Log in as User A with Google already linked
- Navigate to group / Calendar tab
- In the External Calendars panel, open the calendar selector dropdown under the Google connection
- Select a calendar from the list (e.g. "primary")
- Toggle Sync Enabled to on
- Click Save / confirm
Expected: The panel refreshes showing the selected calendar name and sync status

as enabled. Navigating away and returning to the Calendar tab still shows the saved selection.

TC-5-4: Manually sync group events to Google Calendar (AC2)

- Log in as User A with Google linked and sync enabled
- Ensure the group has at least one itinerary event with a title, start time, and end time
- In the External Calendars panel, click Sync Now
Expected: A success message or updated "Last synced" timestamp appears. Opening Google Calendar in a browser shows the itinerary event(s) created in the selected calendar with matching titles and times.

TC-5-5: Re-sync updates existing events instead of creating duplicates (AC8)

- Complete TC-5-4 first so at least one event is synced
- In the group's Itinerary tab, edit the title of the event that was synced (e.g. change it to "Updated Event Title")
- Return to the Calendar tab and click Sync Now again
Expected: Google Calendar shows the event with the updated title. The original event is not duplicated — there is still only one event entry for that itinerary item.

TC-5-6: Attempt to link the same Google account twice (AC4 partial — duplicate prevention)

- Log in as User A who already has Google linked
- Click Connect Google Calendar again and complete the OAuth flow with the same account
Expected: The system prevents creating a duplicate connection. Either the UI hides the connect button when already linked, or the API returns an error and the panel shows "Google Calendar already linked."

TC-5-7: Update sync settings — disable then re-enable sync (AC6)

- Log in as User A with Google linked and sync enabled
- In the External Calendars panel, toggle Sync Enabled to off and save
Expected: Sync status shows as disabled. Click Sync Now (if available), no events should be pushed, or the button is hidden.
- Toggle sync back on and save
Expected: Sync status returns to enabled. Clicking Sync Now pushes events successfully.

TC-5-8: Unlink Google Calendar (AC7)

- Log in as User A with Google linked
- In the External Calendars panel, click Unlink next to the Google connection
- Confirm the action if prompted
Expected: The Google account is removed from the linked calendars list. The panel returns to showing only the Connect buttons. Previously synced events remain in Google Calendar (they are not deleted from the external provider) but future syncs no longer occur.

TC-5-9: Unauthenticated access is denied (AC5)

- Log out of the app
- In a browser, navigate directly to <http://localhost:3000/api/calendar-connections>
Expected: The server responds with a 401 Unauthorized JSON response. No calendar data is returned.

TC-5-10: Expired or revoked credentials show error state (AC9)

- Log in as User A with Google linked
 - In the Google account security settings, revoke access for the BoilerBridge app
 - Return to the app, navigate to the Calendar tab, and click Sync Now
Expected: The panel displays an error message indicating that sync failed due to an authentication or access issue (e.g. "Sync failed: token expired or access revoked"). The last sync timestamp is not updated to a new success time.
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US#6 — Itinerary Map

TC-6-1: View map with valid itinerary locations (AC1)

- Log in as User A
- Navigate to the group then click Map in the tab bar
- Ensure the group has at least one itinerary event with a location string
Expected: The Itinerary Map panel loads. The Google Map renders (no blank grey area). At least one numbered marker is visible on the map at a position corresponding to the event's location. The stop list on the left shows the event title and location.

TC-6-2: Stops display in itinerary order with visual connection (AC2)

- Log in as User A
- Navigate to the group then Map tab
- Ensure the group has at least two itinerary events with different locations, scheduled at different times
Expected: The stop list on the left shows events ordered by start time (earliest first). Each stop has a numbered badge (1, 2, 3...). The markers on the map are numbered to match. An amber polyline connects the markers in sequence from stop 1 to the last stop.

TC-6-3: Click a stop in the list to pan and highlight on the map (AC3)

- Log in as User A, navigate to Map tab with at least two mapped stops visible
- Click on stop #2 in the stop list on the left panel
Expected: The map pans and zooms to the location of stop #2. An info window appears above the marker showing the stop's title, location, and start time. The stop card in the left panel highlights with an amber border.

TC-6-4: Click a marker on the map to see stop details (AC3)

- Log in as User A, navigate to Map tab
- Click directly on a numbered marker on the map
Expected: An info window opens at the marker location showing the stop's title, location address, and start time. If the event has a description, it is also shown below a divider.

TC-6-5: Unauthenticated user cannot load map data (AC4)

- Log out of the app
- Navigate directly to <http://localhost:3000/api/groups/<groupId>/itinerary/map>
Expected: The server responds with 401 Unauthorized. No itinerary stop data is returned.

TC-6-6: Non-member cannot access group map data (AC4)

- Log in as User C (not a member of the group)
- Navigate directly to <http://localhost:3000/api/groups/<groupId>/itinerary/map>
Expected: The server responds with 403 Forbidden. No stop data is returned.

TC-6-7: Map updates after itinerary changes (AC5)

- Log in as User A, navigate to Map tab, note the current stops
- Navigate to the Itinerary tab and add a new event with a location (e.g. "Indianapolis, IN")
- Return to the Map tab
Expected: The new stop appears in the stop list in the correct chronological position. After geocoding completes (brief spinner), a new numbered marker appears on the map at the correct location. The polyline updates to include the new stop.

TC-6-8: Empty state when no itinerary stops have location data (AC6)

- Log in as User A in a group where all itinerary events have no location field, or in a group with no itinerary events at all
- Navigate to Map tab
Expected: No map is rendered. Instead, a centered empty state is shown with a map pin icon and a message such as "No itinerary events yet" or "No stops could be mapped. Add a location to your itinerary events to see them here." No blank or broken map area is visible.

TC-6-9: Stops with no location show as unmapped in the list (AC6 partial)

- Log in as User A in a group that has a mix of events, some with locations, some without
- Navigate to Map tab
Expected: Events without a location appear in the stop list with a greyed-out appearance and a "-" badge instead of a number. They do not appear as markers on the map. A note at the bottom of the list indicates how many locations could not be geocoded.

US#7 — Writing Reviews

Prerequisites

- App running at <http://localhost:3000>
- User A has an account and has created an activity

TC-7-1: Reviews Populate on Load (AC1)

- Log in as User A
- Open activities
- Click on activity listed

Expected: Activity review page loads and populates reviews.

TC-7-2: Writing a review (AC2)

- Log in as User A
- Open activities
- Click on activity listed
- Write and submit a review

Expected: The review is sent to the server and populates on the activity page user A is on and persists on reload.

TC-7-3: Deleting a review (AC3)

- Log in as User A
- Open activities
- Click on activity listed
- Write and submit a review
- Press delete button on review

Expected: The review is deleted on the activity page user A is on and persists on reload.

US#8 — Itinerary Link Sharing

Prerequisites

- App running at <http://localhost:3000>
- User A owns itinerary with multiple days / activities
- User B is non-connected account

TC-8-1: Getting share link (AC1)

- Log in as User A
- Open group
- Click on “Copy share link”, ensuring sharing is enabled with the toggle beside
- User B opens link for shared itinerary

Expected: User A's group travel plans and itinerary populate on page served by share link.

TC-8-2: Toggling sharing (AC3)

- Omitted due to only setting being a sharing toggle and not needing a full popup UI for itinerary sharing settings.

TC-8-3: Toggling sharing (AC2)

- Log in as User A
- Open group
- Toggle the setting beside the copy share link button

Expected: User A's group travel plans and itinerary should not populate on the page served by share link.

US#9 — 10-Day Weather Forecast

Prerequisites

- App running at <http://localhost:3000>
- User A owns itinerary with multiple days / activities

TC-9-1: View forecast (AC1)

- Log in as User A
- Open itinerary
- Scroll down

Expected: Panel with 10-Day weather forecast is present and populated.

TC-9-2: View forecast origin (AC2)

- Log in as User A
- Open itinerary
- Scroll down
- Click on either link after "From:" in header of forecast

Expected: User A is taken to the link for weather data origins

TC-9-3: Change forecast location (AC3)

- Log in as User A
- Open all trips
- Click change preferences on the trip created previously
- Change trip location
- Go to that trips group page
- Scroll down

Expected: 10-Day weather forecast now shows weather for new location.

US#10 — In-App Trip Notifications

Prerequisites

- App running at <http://localhost:3000>
- User A owns itinerary with multiple days / activities
- User B is non-owner account and is part of User A's group

TC-10-1: Notifications Populated (AC1)

- Log in as User A

- Open notification bell in top right

Expected: Notifications from while away from site have populated.

TC-10-2: Notifications Populated (AC2)

- Log in as User A
- Click settings tab
- Edit in-app trip notification preference
- Click save
- Reload page

Expected: Setting change reflects immediately and persists on reload.

TC-10-3: Notifications Populated (AC3)

- Log in as User A
- Log in as User B in another window
- Edit trip preferences or change/generate itinerary
- Open notification bell on User B account

Expected: A notification should populate on User B's client regarding the update.

US#15 — Public Itinerary Feed

Prerequisites

- App running at <http://localhost:3000>
- Two test accounts: User A (owner), User B (different user)
- User A has at least one saved itinerary
- User B has no edit access to User A itineraries
- Public Feed page available in navigation

TC-15-1: Publish an itinerary (AC1)

- Log in as User A
- Navigate to My Trips / My Itineraries
- Open an existing itinerary
- Click Publish

Expected: Success message shown. Itinerary now appears in Public Feed.

TC-15-2: Verify itinerary appears in public feed (AC1)

- Stay logged in as User A or log in as User B
- Navigate to Public Feed

Expected: Published itinerary card is visible with preview details such as destination, duration, highlights.

TC-15-3: Open public itinerary details (AC2)

- From Public Feed, click View Details on a public itinerary

Expected: Full itinerary opens with schedule/details visible. No edit controls shown.

TC-15-4: Unauthorized user cannot edit public itinerary (AC2)

- Log in as User B
- Open User A's public itinerary

- Try direct edit URL or API request

Expected: Edit blocked. UI hides controls or API returns 403 Forbidden.

TC-15-5: Views increment on open

- Open same itinerary details page multiple times

Expected: View count increases in database or admin/dev tools.

TC-15-6: Sorting by latest

- Publish multiple itineraries at different times
- Open Public Feed sorted by Latest

Expected: Most recently published itinerary appears first.

TC-15-7: Unpublish itinerary (AC3)

- Log in as User A
- Open own published itinerary
- Click Unpublish
- Refresh Public Feed

Expected: Itinerary no longer appears in feed.

TC-15-8: Only owner can toggle visibility

- Log in as User B
- Try publish/unpublish User A itinerary via API/UI

Expected: Request rejected with 403 Forbidden.

US#16 — Partial Itinerary Editing

Prerequisites

- App running at <http://localhost:3000>
- User A owns itinerary with multiple days / activities
- User B is non-owner account
- Itinerary contains at least 3 editable activities

TC-16-1: Edit one activity only (AC1)

- Log in as User A
- Open itinerary
- Click Edit on Activity #2 only
- Change title/time/location
- Save

Expected: Only Activity #2 updates. Other activities remain unchanged.

TC-16-2: Refresh persistence check (AC2)

- After saving changes, refresh page

Expected: Updated activity still shows modified values.

TC-16-3: Edit one day section only

- Open Day 2 section

- Modify notes or replace one activity

Expected: Only Day 2 changes. Day 1 and Day 3 untouched.

TC-16-4: Optimistic UI behavior

- Edit one activity and click Save

Expected: UI updates immediately without full page reload.

TC-16-5: Unauthorized user blocked (AC4)

- Log in as User B
- Attempt to edit User A itinerary section

Expected: Edit denied. API returns 403 Forbidden.

TC-16-6: Concurrent edits on different sections (AC3)

- User A opens itinerary in Browser 1
- User A or User B collaborator opens same itinerary in Browser 2
- Browser 1 edits Activity #1
- Browser 2 edits Activity #3

Expected: Both changes saved. Unrelated sections not overwritten.

TC-16-7: Same section conflict handling

- Open same activity in two browsers
- Save different edits from both sessions

Expected: System follows defined conflict rule (latest save wins or version conflict message).

TC-16-8: Invalid partial payload rejected

- Send PATCH request with missing activity ID or bad values

Expected: API returns validation error 400. No unintended changes saved.